MEDICAL EXPERT REPORT

FROM: Dr. Mark Ettinger
TO: Sorrels Law
DATE: September 11, 2024
RE: *Chad Alan Warner and Elizabeth Warner v. Dr. Matthew John Maxwell*

This medical expert report evaluates the medical care rendered to Mr. Chad Alan Warner that is the subject of the above-referenced lawsuit. This report is based on a review of the following information:

- The medical records of Mr. Warner; and
- Plaintiff's Original Petition in this case.

The opinions contained in this report are stated to a reasonable degree of medical certainty, i.e., based on a "more likely than not" standard, unless otherwise stated. Furthermore, I use the following definitions:

1. "Ordinary care" means that degree of care that an anesthesiologist of ordinary prudence would use under the same or similar circumstances.
2. "Negligence" means failure to use ordinary care that an anesthesiologist of ordinary prudence would have done under the same or similar circumstances or doing that which a anesthesiologist of ordinary prudence would not have done under the same or similar circumstances.
3. "Proximate cause" means a cause that was a substantial factor in bringing about an injury, and without which cause such injury would not have occurred. To be a proximate cause, the act or omission complained of must be such that a person using ordinary care would have foreseen that the injury, or a similar injury, might reasonably result from said act or omission. There may be more than one proximate cause of an injury.

QUALIFICATIONS AND EXPERIENCE

I have been practicing medicine in the State of Texas as an anesthesiologist and critical care physician since 2011. I graduated from the LSU School of Medicine in New Orleans and did residency training in both neurological surgery and anesthesiology. I completed my anesthesiology and critical care medicine residency at The Johns Hopkins Hospital in 2011. I am Board Certified in Anesthesiology and I currently practice at a very large volume, level 2 trauma center in a suburb of Fort Worth, Texas. I perform between 1000-1500 anesthetics per year in patients aged 1 to 100+. Our patient population ranges from very healthy to extremely ill and I routinely take care of patients with cardiac dysfunction. I serve on multiple committees involved in case review, including as the Chairman of the hospital's Medical Ethics Committee and the Peer Review Committee. I am very familiar with all aspects of this particular case, as I have provided anesthesia for countless cardiac ablation procedures, frequently using this same anesthetic technique. My facility does between 200 and 400 of these procedures each year. I am very familiar with the standards of care in patient selection, anesthetic administration and post-operative care for these types of cases. I am well-qualified to provide expert testimony on the anesthetic management of this case.

SUMMARY OF FACTS

On November 14, 2023 Plaintiff Chad Alan Warner visited his cardiologist, Dr. Rajesh Ramineni, with complaints of shortness of breath and chest tightness. During the previous two weeks, Mr. Warner suffered from pneumonia that had been treated with antibiotics. Dr. Ramineni found that Mr. Warner was suffering from atrial flutter with rapid ventricular rate—a condition that caused Mr. Warner's heart to beat rapidly and irregularly—and referred Mr. Warner to the emergency department at Houston Methodist The Woodlands Hospital ("HMTW"). There, follow-up electrocardiograms confirmed that Mr. Warner had atrial flutter with rapid ventricular rate. The history and physical note signed by emergency medicine Dr. Christopher Fedoruk indicates that the patient had an enlarged heart and fluid around his heart. Mr. Warner's past medical history included being morbidly obese with a body mass index of 41. Mr. Warner tested positive for respiratory syncytial virus. Mr. Warner's lab tests showed that he had abnormally high levels of troponin-T and NT-proBNP which suggest that Mr. Warner's heart had suffered from some injury and/or failure.

In the emergency department, Mr. Warner was given the maximum rate of an intravenous infusion of diltiazem, a heart rate lowering drug, but his heart rate remained high. Dr. Rajesh Venkataraman, a cardiologist who specializes in electrophysiology, was consulted and scheduled Mr. Warner to undergo a cardiac ablation procedure the following day to treat the atrial flutter. A transthoracic echocardiogram was ordered by nurse practitioner John Hustedt to evaluate Mr. Warner's heart but was never performed before the cardiac ablation. Early in the morning of November 15, 2023, Mr. Warner was transferred to the intensive care unit. A CT angiogram/ pulmonary vein mapping imaging exam performed at 1258 showed that Mr. Warner had four-chamber cardiomegaly. Four-chamber cardiomegaly is a condition in which all four chambers of the heart are enlarged.

On November 15, 2023 at 1335 Mr. Warner was transported from the intensive care unit to the cardiac catheterization lab to undergo the cardiac ablation under general anesthesia. His anesthesiologist was Dr. Matthew John Maxwell who was assisted by certified registered nurse anesthetist Sarah Catherine Coppin. At 1358, before Mr. Warner was given any anesthetic drugs, his heart rate was 133, blood pressure was 115/75 as read by a blood pressure cuff, oxygen saturation was 100%. Induction of anesthesia at 1401, the anesthesia team gave Mr. Warner 280 milligrams (mg) of propofol and 100 micrograms (mcg) of fentanyl intravenously (IV). Immediately Mr. Warner went into asystole (i.e., his heart stopped beating). Consistent with this event, the end-tidal carbon dioxide ("$ETCO_2$"), an indicator of cardiac output, dropped to 12 mm Hg, an abnormally low level. A code blue was called and chest compressions were begun. The anesthesia record indicates that the code blue was called at 1403 whereas the nursing record indicates that the code blue was called at 1406.

From 1402 to 1418 Mr. Warner's heart remained stopped, and from 1402 to 1419 Mr. Warner did not have a measured blood pressure. These vital signs indicate that Mr. Warner's brain during this time period did not receive adequate perfusion of blood or oxygen. Once a spontaneous heartbeat returned at around 1418, Dr. Venkataraman performed an intracardiac echocardiogram which revealed that the heart was at a near standstill. Per the anesthesia record chest compressions were then restarted at 1434 and halted at some time later that is not documented in the medical record.

Because the heart was not pumping sufficient blood, an Impella ventricular assist device was placed by Dr. Agusala at 1421 to help Mr. Warner's heart pump blood. To further improve oxygenation of the blood, Mr. Warner was then taken to an operating room and placed on extracorporeal membrane oxygenation ("ECMO"). After these lifesaving procedures were performed, Mr. Warner was transferred from the cardiac catheterization lab to the ICU at 1821. He was then transferred from HMTW to Houston Methodist Hospital in Houston, Texas on November 17, 2023.

Since the catastrophic events of November 15, 2023, Mr. Warner has not regained full consciousness. He has been diagnosed with severe brain damage and a disorder of consciousness and is in a vegetative state. He requires a tracheostomy tube to breathe, a feeding tube to receive nutrition, and around-the-clock nursing care.

CLINICAL ANALYSIS AND STANDARD OF CARE

Generally, the standard of care requires an anesthesiologist to evaluate a patient properly before surgery, use the proper drugs for a given patient and surgery, and use appropriate monitoring for the patient and surgery. The proper dugs and monitoring depend on factors related to both the patient and surgery. Patient-related factors include the patient's current medical conditions, past medical history, physiological characteristics, and others. Surgery-related factors include the surgery's potential to impact any organ system, the degree of expected surgical stimulation, degree of paralysis needed, potential for blood loss, and other related factors.

Proper evaluation of a patient prior to surgery in a case like this would include a thorough evaluation of the patient's cardiac function. The quickest, most non-invasive and effective way to evaluate cardiac function before surgery is to perform an echocardiogram. This is an ultrasound of the heart that provides a very detailed picture of the anatomy of the heart and its overall function. This will show if there are any abnormalities with the chambers or valves of the heart and it will show the beating of the heart in real time allowing one to assess the quality of function of the heart. This study will generate a number known as the ejection fraction, which provides a measure of how strong the heart is beating. A normal EF is usually 50-70%. When patients have an EF that is considerably low, the standard of care is to change the anesthetic plan to a technique that accommodates for the depressed heart function. Given this patient's symptoms, known arrhythmia, and lab values showing heart damage, the standard of care would include obtaining an echocardiogram before administering anesthesia. An echocardiogram was ordered, indicating recognition of its need for this case. However, it was never performed. Another important aspect of the preoperative evaluation would be to assess a patient's exercise tolerance. One common way of performing this inquiry is by asking the patient if he can walk two flights of stairs without getting short of breath or having chest pain. A negative answer indicates that the patient may have some form of cardiovascular disease and would warrant a further inquiry into the exact symptoms that the physical exertion triggers. If the anesthesiologist determines that the patient's symptoms are caused by cardiovascular disease and are severe, the anesthesiologist should take precautionary measures, such as avoiding cardiac depressant drugs and placing an arterial line before anesthesia induction. This inquiry was especially important considering Mr. Warner's acute cardiac condition that required admission to the intensive care unit.

The standard of care in the preoperative evaluation of a patient prior to surgery also includes optimizing that patient's medical conditions in preparation for anesthesia. In this particular case, the most important preoperative factor requiring optimization is the patient's elevated heart rate. In order to function properly, the heart needs an adequate time to refill after each beat. An elevated heart rate reduces the time available for the heart to refill and an extremely high heart rate can result in the heart having little to no blood available to pump during each contraction. This greatly reduces the blood pressure and oxygen perfusion to the brain. In addition, the heart itself needs oxygenated blood, and it relies on the refilling of the heart to supply that blood. In patients with normal cardiac function, the heart can usually weather the storm through an elevated heart rate for an extended period of time. But in a patient with cardiac dysfunction, this can off lead to a lack of blood and oxygen to the heart muscle itself and result in a myocardial infarction, arrhythmia, or complete arrest of the heart, as seen in this case. Since the ablation procedure was not an emergency procedure, the standard of care would be to optimize the heart rate prior to performing the ablation. This would typically be done by giving medications and, if those don't work, performing a cardio version or "shocking the heart" to convert the rhythm back to a normal rhythm and normalize the heart rate. These efforts were not made in this case—the patient was given a drug that slows the heart, diltiazem, which was not successful in lowering the heart rate. No other drugs were given and a cardio version was not performed.

Furthermore, the standard of care requires an anesthesiologist to recognize when certain anesthesia drugs alone or in combination pose a risk of causing an adverse physiological impact on a patient and how to mitigate that risk. In this case Mr. Warner was known to have acute atrial flutter with a rapid ventricular rate that ranged from about 120 to 160 beats per minute. When the heart is in atrial flutter the usual timing and coordination of atrial and ventricular contractions is disturbed, resulting in the reduction of cardiac output. The rapid ventricular rate further impaired the cardiac output by not allowing enough time between cardiac contractions to enable the heart to completely fill up with blood. The atrial flutter with rapid ventricular rate combined with the elevated levels of troponin T and NT-proBNP indicate that Mr. Warner's heart muscle had been damaged, and his cardiac function was severely impaired.

Propofol is an intravenous anesthetic drug that causes the loss of consciousness within seconds of administration. One of the side effects of propofol, however, is that it depresses the contractility of heart muscle. Contractility is the strength of contraction of a muscle. Another side effect of propofol is that it lowers blood pressure (by reducing systemic vascular resistance). In general blood pressure must be maintained to ensure that the heart receives a sufficient supply of blood and oxygen. A drop in blood pressure below a critical level causes the heart to receive an insufficient amount of oxygen. Once a critical drop in blood pressure occurs, after some time the heart muscle will lose its ability to contract and will arrest. The time required for a cardiac arrest to occur varies from patient to patient, depending in part on the health of the patient's heart, but it can be seconds. Therefore, one of the goals during anesthesia is to maintain a mean arterial pressure ("MAP") of at least 60 millimeters of mercury ("mm Hg"). Mean blood pressure is calculated by the following formula:

MAP = (2/3 x diastolic blood pressure) + (1/3 x systolic pressure)

Fentanyl can also cause slowing of the heart rate and reduction of blood pressure. Thus, the combination of propofol and fentanyl, can cause a severe reduction of blood pressure, particularly when given simultaneously.

Patients with existing cardiac dysfunction are sensitive to drugs that depress the cardiovascular system which can exacerbate the cardiac dysfunction. For such patients even a slight amount of further cardiovascular depression can trigger a downward spiral that can have fatal consequences. Thus, in patients, such as Mr. Warner, who have existing cardiac dysfunction, the standard of care requires inducing anesthesia in such a way that avoids further cardiac depression. There are various anesthetic techniques to avoid causing cardiac depression. One technique is to use etomidate, an anesthetic induction drug commonly used on patients with cardiac disease, because etomidate has the lowest cardiovascular depressant effect among the available anesthetic induction agents. Another technique, if the anesthesiologist chooses to use drugs that have significant cardiovascular depressant effect, is to administer the drugs in small boluses over several minutes while closely monitoring the heart rate and blood pressure. This technique avoids sudden and significant drops in heart rate and blood pressure, and allows the anesthesiologist time to respond with drugs that counteract any drops of the heart rate or blood pressure.

Further, because patients with depressed cardiac function are susceptible to sudden and significant changes of their blood pressure, an arterial line placed before anesthesia induction is required by the standard of care. The arterial line gives an instantaneous and continuous reading of the blood pressure in real time. In contrast, a non-invasive blood pressure cuff, which was used in this case, is not able to give an instantaneous continuous reading of the blood pressure. A non-invasive blood pressure cuff typically takes thirty to sixty seconds to measure the blood pressure, and can give an inaccurate reading if the blood pressure is too low or the blood pressure becomes non-pulsatile. Using an arterial line to know the blood pressure at all times enables the anesthesiologist is to respond to any changes in the blood pressure immediately.

The following table sets forth the timeline of events that occurred.

Time	Heart Rate/	BP	O2 Sat	ET CO2	Event/Action Taken by Anesthesia Team
135	143				
135	141	118/86	100%		
135	141		100%		
1358	135	115/74	100%		Propofol 200 mg, fentanyl 100 mcg, and rocuronium 10 given.
135	133		99%		Succinylcholine 180 mg given
140	128		92%		Propofol 80 mg given. Phenylephrine 200 mcg given
140	122				
140	111			28	Intubation.
140	0		77%	12	Phenylephrine 200 mcg given.
140	16		77%	8	Code blue begun. Epi 1 mg given.
140	23		90%	24	
140	29		100%	10	
1406	Blank pulse		100%	17	Epi 1 mg given.
140	8				
140	49			15	
140	58			16	
141	0			9	
141	0				
141	0		67%		Epi 1 mg given.
141	0	65/45	59%	9	
141			61%	2	
141			76%	3	
141			69%	0	
141	0		67%		Epi 1 mg given.

141	192		71%	0	
141	98	111/47		19	
142	100		66%	21	
142	30		100%	50	
142					
142					
142					
142		161/81			

After anesthesia was induced at 1358, Mr. Warner suffered a cardiac arrest at 1402. According to the anesthesia record, at 1403 a code blue was called, chest compressions were started, and epinephrine 1 mg was given. As the cardiac arrest continued, a second dose of epinephrine 1 mg was given at 1406. The third dose of epinephrine 1 mg was given at 1412. According to Advanced Cardiovascular Life Support ("ACLS") protocol, during a cardiac arrest epinephrine 1 mg should be given every three to five minutes while chest compressions are given.

To summarize, the standard of care in this case required the anesthesiologist to:

1) Perform a proper and thorough preoperative evaluation prior to surgery, which in this case would have included assessing exercise tolerance and performing and evaluating an echocardiogram prior to administering anesthesia;

2) Optimizing the patient's cardiac condition prior to surgery, which in this case would have involved properly controlling the rate of the heart prior to induction of anesthesia;

3) Adjusting the anesthetic plan to compensate for the patient's depressed cardiac function. This would include:

   1) avoid giving drugs that will cause excessive depression of the patient's cardiovascular system;

2) Adjust the normal dosing of drugs used to account for depressed cardiac function;

3) Induce anesthesia in a slow, controlled fashion while closely monitoring the patient's vital signs;

4) Monitor the patient's blood pressure in real time by using an arterial line placed prior to the induction of anesthesia

4) Follow ACLS protocol during a cardiac arrest by giving epinephrine at least once every 3 to 5 minutes.

BREACH OF STANDARD OF CARE

Dr. Matthews breached the standard of care by:

1) Not performing an adequate preoperative evaluation, to include both assessment of exercise tolerance and evaluation of heart function by echocardiogram;

2) Not optimizing the patient prior to proceeding for surgery. Specifically in this case, failing to take additional efforts to control the patient's heart rate before the procedure. This would have included administering other drugs to control heart rate or performing an electrical cardioversion;

3) Choosing a primary anesthetic that is severely depressing to heart function and giving that anesthetic in a larger than indicated dose. The standard of care in this case would be to either use an alternative drug that does not significantly depress the heart, such as etomidate, or to use propofol in a much lower dose in conjunction with other agents.

4) Failing to induce anesthesia in a slow, controlled manner while closely monitoring the patient's vital signs;

5) Failing to use a pre-induction arterial line to more closely monitor Mr. Warner's blood pressure; and

6) Failing to follow ACLS protocol during the cardiac arrest by not giving epinephrine at least once every three to five minutes.

CAUSATION

By committing the breaches of the standard of care enumerated above, Dr. Matthews caused Mr. Warner to suffer two cardiac arrests, the first arrest occurring between 1402 and 1419 and the second arrest occurring after 1434 for an undocumented period of time. Dr. Matthew's failure to assess the cardiac function by echocardiogram prior to the procedure resulted in him choosing an anesthetic plan that did not account for this patient's likely cardiac dysfunction. Dr. Matthew's decision to proceed with surgery without first optimizing the patient's condition, specifically his heart rate, created a scenario in which any significant reduction in blood pressure or supply of oxygen to the heart would result in oxygen deprivation to the heart. During the induction of anesthesia, the simultaneous and rapid administration of propofol 280 mg and fentanyl 100 mcg caused severe depression of the heart's contractility and reduction of the blood pressure. The blood pressure reduction in turn caused a critical reduction of the supply of blood and oxygen to the heart, which was already compromised and under high demand due to tachycardia, causing it to stop beating. At this point, the primary goal is to get the heart restarted, which requires following ACLS protocols. Those protocols were not properly followed during resuscitation. Once the heart arrested, blood flow to the brain ceased, depriving the brain of oxygen. Brain tissue becomes permanently damaged after it is deprived of oxygen for around five minutes. In this case Mr. Warner's brain was deprived of oxygen for at least seventeen minutes. As a result, Mr. Warner suffered from an anoxic brain injury from which he has not recovered. Today Mr. Warner suffers from diffuse encephalopathy and disorder of consciousness. In addition, his heart suffered from acute heart failure leading to now-chronic heart failure. In my expert opinion, I can say beyond a reasonable degree of medical certainty that the breaches of the standard of care mentioned herein were the direct cause of Mr. Warner's injury.

/s/ Dr. Mark Ettinger